t-test(for Dependent Sample)

Steps for Significance Testing

1. Set alpha (p level).
2. State hypotheses, Null and Alternative.
3. Calculate the test statistic (sample value).
4. Find the critical value of the statistic.
5. State the decision rule.
6. State the conclusion.

Types

1. The ***independent-sample t test*** is used to compare two groups' scores on the same variable. For example, it could be used to compare the salaries of dentists and physicians to evaluate whether there is a difference in their salaries.

2. The ***paired-sample t test*** is used to compare the means of two variables within a single group. For example, it could be used to see if there is a statistically significant difference between starting salaries and current salaries among the general physicians in an organization.

Dependent Samples t-tests

Dependent Samples *t*-test

- Used when we have dependent samples – matched, paired or tied somehow
  - Repeated measures
  - Brother & sister, husband & wife
  - Left hand, right hand, etc.
- Useful to control individual differences. Can result in more powerful test than independent samples *t*-test.

Dependent Samples *t*

Formulas:

$$t_{\overline{X}_D} = \frac{\overline{D}}{SE_{diff}}$$

t is the difference in means over a standard error.

$$SE_{diff} = \frac{SD_D}{\sqrt{n_{pairs}}}$$

The standard error is found by finding the difference between each pair of observations. The standard deviation of these difference is SD_D. Divide SD_D by sqrt (number of pairs) to get SE_{diff}.

Another way to write the formula

$$t_{\overline{X}_D} = \frac{\overline{D}}{SD_D \Big/ \sqrt{n_{pairs}}}$$

Dependent Samples *t* example

Person	Painfree (time in sec)	Placebo	Difference
1	60	55	5
2	35	20	15
3	70	60	10
4	50	45	5
5	60	60	0
M	55	48	7
SD	13.23	16.81	5.70

Dependent Samples *t* Example (2)

1. Set alpha = .05
2. Null hypothesis: H_0: $\mu_1 = \mu_2$.
   Alternative is H_1: $\mu_1 \neq \mu_2$.
3. Calculate the test statistic:

$$SE_{diff} = \frac{SD}{\sqrt{n_{pairs}}} = \frac{5.70}{\sqrt{5}} = 2.55$$

$$t = \frac{\overline{D}}{SE_{diff}} = \frac{55 - 48}{2.55} = \frac{7}{2.55} = 2.75$$

Dependent Samples t Example (3)

4. Determine the critical value of t.
   Alpha =.05, tails=2
   df = N(pairs)-1 =5-1=4.
   Critical value is 2.776
5. Decision rule: is absolute value of sample value larger than critical value?
6. Conclusion. Not (quite) significant. Painfree does <u>not</u> have an effect.